Original Article

Journal of INTERNAL MEDICINE

doi: 10.1111/j.1365-2796.2011.02470.x

Hereditary lysozyme amyloidosis – phenotypic heterogeneity and the role of solid organ transplantation

P. T. Sattianayagam, S. D. J. Gibbs, D. Rowczenio, J. H. Pinney, A. D. Wechalekar, J. A. Gilbertson, P. N. Hawkins, H. J. Lachmann & J. D. Gillmore

National Amyloidosis Centre, Centre for Amyloidosis & Acute Phase Proteins, Department of Medicine, Royal Free Campus, University College London Medical School, London, UK

Abstract. Sattianayagam PT, Gibbs SDJ, Rowczenio D, Pinney JH, Wechalekar AD, Gilbertson JA, Hawkins PN, Lachmann HJ, Gillmore JD (University College London Medical School, London, UK). Hereditary lysozyme amyloidosis – phenotypic heterogeneity and the role of solid organ transplantation. *J Intern Med* 2012; **272**: 36–44.

Objectives. Lysozyme amyloidosis (ALys) is a form of hereditary systemic non-neuropathic amyloidosis, which is inherited in an autosomal dominant fashion. Lysozyme, which is the amyloidogenic precursor protein in ALys, is a ubiquitous bacteriolytic enzyme synthesized by hepatocytes, polymorphs and macrophages. The aim of this study is to describe the phenotype and outcome of patients with ALys including the role of solid organ transplantation.

Design. Retrospective evaluation of patients with ALys.

Setting. UK National Amyloidosis Centre.

Patients. All 16 patients with ALys followed at the centre.

Results. A family history of amyloidosis was present in every affected individual. Although the phenotype was broadly similar amongst those from the same kindred, there were marked phenotypic differences between kindreds who possessed the same amyloidogenic mutation. Symptomatic gastrointestinal (GI) amyloid was prevalent, and macroscopically visible amyloidotic lesions were present in nine of 10 patients who underwent GI endoscopy. All symptomatic ALys individuals had hepatic amyloid. Four patients received orthotopic liver transplants (OLT), three for spontaneous hepatic rupture and one case, who had extensive hepatic amyloid and a strong family history of hepatic rupture, pre-emptively. All of the liver grafts were functioning at censor 1.7, 5.8, 9.0 and 11.0 years after OLT. Five patients had progressive amyloidotic renal dysfunction culminating in end-stage renal failure, three of whom underwent renal transplantation (RTx). There was no evidence of renal allograft dysfunction at censor 6.6, 1.8 and 0.8 years after RTx.

Conclusions. Lysozyme amyloidosis is a disease of the GI tract, liver and kidneys, which has a slow natural history. There was a clear family history in all cases within this cohort, demonstrating a high clinical penetrance in the presence of an amyloidogenic lysozyme mutation. There is currently no amyloid-specific therapy for the condition which is managed symptomatically. OLT and RTx appear to be successful treatments for patients with liver rupture or end-stage renal disease, respectively, with excellent outcomes in terms of medium-term graft function and patient survival.

Keywords: amyloidosis, gastrointestinal, kidney, liver, lysozyme, transplantation.

Introduction

Systemic amyloidosis is a disease characterized by deposition of protein in an abnormal fibrillary conformation which causes disruption of tissue structure and function. Hereditary non-neuropathic systemic amyloidosis was first described in 1932 by Ostertag [1] and represents a group of diseases inherited in an autosomal dominant manner. A number of causative mutations in genes encoding amyloidogenic variant precursor proteins have been identified: apolipoprotein AI (16 mutations), apolipoprotein AII (four mutations), fibrinogen Aα-chain (nine mutations) and lysozyme (six mutations). The phenotype of these conditions is variable in terms of penetrance, age of onset, organ involvement, rate of progression and prognosis, not only within a given amyloidosis type but even between those with the same mutation.

 © 2011 The Association for the Publication of the Journal of Internal Medicine

Lysozyme amyloidosis was first described in 1993 by Pepys *et al.* [2] and *in vitro* studies with lysozyme have provided seminal insights into amyloid fibrillogenesis. The phenotype of ALys is heterogeneous and includes gastrointestinal (GI) symptoms, hepatic rupture, sicca syndrome, petechiae and purpura, renal failure and lymphadenopathy. Lysozyme is a bacteriolytic enzyme that is synthesized in GI tract macrophages and in hepatocytes. Six different mutations of the lysozyme gene have been reported to encode amyloidogenic variants of the protein [2–6], and a non-amyloidogenic polymorphism has also been reported [7].

The phenotype and natural history of ALys have largely been reported in case reports or kindred analyses. In this series, which is the largest reported to date and which includes prolonged patient follow-up, there is characterization of the clinical presentation and natural history of ALys, including the role of solid organ transplantation in this disease.

Aims and methods

Patients

Sixteen patients who were followed prospectively at the National Amyloidosis Centre (NAC) between 1988 and 2010 were identified retrospectively from the centre database. Diagnosis of ALys required identification of amyloid deposits within tissues (by histology and ^{123}I-serum amyloid P component (SAP) scintigraphy in 13 cases and by ^{123}I-SAP scintigraphy alone in three cases) in combination with the presence of a mutation in the lysozyme gene, corroborated by a family history of amyloidosis and absence of raised inflammatory markers or a clonal disorder.

Patients attended the NAC for their initial diagnostic evaluation and were followed up at regular (6 monthly to yearly) intervals for evaluation of organ function including detailed echocardiographic evaluation of the heart and monitoring of whole body amyloid load by serial ^{123}I-SAP scintigraphy. Additional investigations were undertaken when clinically indicated. Amongst cases where organ failure supervened, listing for organ transplantation was according to local transplant centre protocols.

The medical care described here was performed with informed consent from each patient in accordance with the Declaration of Helsinki. Institutional review board approval for the study was obtained from the Royal Free Hospital Ethics committee.

Histology and immunohistochemistry

Sections from formalin fixed paraffin-embedded biopsies were stained for amyloid with Congo red and viewed under cross-polarized light [8]. Immunohistochemical staining of the amyloid deposits was performed using monoclonal antibodies reactive with serum amyloid A protein (SAA), kappa and lambda immunoglobulin light chains, transthyretin, fibrinogen, apolipoprotein AI and lysozyme, as previously described [9].

DNA analysis

Genomic DNA was extracted from whole blood treated with ethylenediaminetetraacetic acid as previously described [10]. Exons 2 and 4 of the lysozyme gene (GenBank accession no NM_000239) were amplified by polymerase chain reaction assay and analysed by automated sequencing, as described previously [2].

Radiolabelled serum amyloid P component (SAP) scintigraphy

Whole body anterior and posterior scintigraphic imaging was undertaken 6 or 24 h after administration of ^{123}I-labelled SAP, as previously described [11]. Labelled SAP studies were interpreted by a panel of physicians with experience of over 10 000 SAP scans.

Results

Patients

Sixteen patients (five men) with ALys were identified and all were Caucasian. Median (range) age of development of amyloid-related symptoms was 25 (9–70) years, delay from symptom onset to diagnosis of amyloidosis was 2.3 (0–29.3) years, and median (range) age of presentation to our unit was 45 (18–72) years. Patient 8 (Table 1) was referred for genetic screening and was the only asymptomatic case in the series and remained so until her death aged 72 years. Her sister (Patient 7, Table 1) presented with renal amyloidosis aged 66 years.

The clinical heterogeneity of ALys is highlighted in Table 1. Presentation was most commonly with sicca syndrome, from presumed amyloidotic salivary gland infiltration (six cases), or with renal dysfunction (five cases). Rupture or haematoma of the liver was the presenting feature in two cases, and two further cases presented with GI disturbance. One case was asymptomatic and detected by family screening. No patient

© 2011 The Association for the Publication of the Journal of Internal Medicine

13652796, 2012, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2796.2011.02470.x by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [26/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1 *Clinical Features and outcome amongst patients with hereditary lysozyme amyloidosis*

Patient number (gender)	Family number	Lysozyme variant	Mode (age) of presentation	Age at diagnosis (organ biopsy)	Additional clinical features	Gastrointestinal symptoms/ upper gastrointestinal tract endoscopic findings	Dead (D) or Alive (A)/time from diagnosis to death or censor (years)
1 (F)	1	D67H	Sicca syndrome (11)	28 (GI)	B	Nausea and vomiting/Oesophageal and gastric nodules	A/2.5
2 (M)	1	D67H	Sicca syndrome (25)	54 (GI)	–	Dyspepsia/extensive upper gastrointestinal tract ulceration	A/7.6
3 (F)	1	D67H	Sicca syndrome (11)	27 (GI)	–	Dyspepsia/gastritis	A/21.6
4 (F)	2	D67H	Liver rupture (24)	24 (Liver)	S	Haemorrhage/haemorrhagic gastritis and ulceration	A/5.8
5 (M)	2	D67H	GI haemorrhage (9)	9 (GI)	L, LN	Haemorrhage/multiple bleeding polyps from oesophagus to duodenum, haemorrhagic gastritis	D/17.9
6 (M)	2	D67H	Liver haematoma (34)	47 (GI)	R, B	Haemorrhage/peptic ulcer, friable lesions	A/15.5
7 (F)	3	D67H	Renal dysfunction (67)	67 (Kidney)	–		D/4.8
8 (F)	3	D67H	None/FHx	72 (–)	–		D/2.0
9 (M)	4	D67H	Sicca syndrome (20)	22 (Kidney)	R, S	Dyspepsia and diarrhoea/normal	A/29.6
10 (F)	4	D67H	Renal dysfunction (51)	51 (Kidney)	–	Anaemia/–	A/12.1
11 (F)	4	D67H	Renal dysfunction (51)	51 (–)	S	Haemorrhage, diarrhoea/–	D/10.9
12 (F)	5	D67H	Sicca syndrome (13)	34 (Liver)	L, S, B	Abdominal pain/peptic ulcer, friable lesions	A/9.1
13 (F)	6	I56T	Renal dysfunction (70)	71 (–)	B		D/5.8
14 (F)	7	W64R	Sicca syndrome (13)	34 (GI)	S, LN	Haemorrhage/peptic ulcer, friable lesions	A/6.6
15 (F)	7	W64R	GI haemorrhage (52)	52 (GI)	S	Haemorrhage/peptic ulcer, friable lesions	A/10.5
16 (M)	8	D67G	Renal dysfunction (29)	32 (Spleen)	S, B, SP		A/1.8

S, sicca syndrome; GI, gastrointestinal; B, bruising; FHx, family history, LN, lymphadenopathy; SP, splenic rupture; R, renal dysfunction; L, liver rupture.

13652796, 2012, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2796.2011.02470.x by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [26/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

had symptoms or investigations to suggest involvement of the heart by amyloid.

Histology

Amyloid was confirmed by histology in 13 patients, from biopsies of the GI tract (seven cases), kidney (three cases), explanted liver after rupture (two cases) and spleen after rupture (one case). Positive immunohistochemical staining with antibodies against lysozyme and negative staining with all other antibodies confirmed the amyloid fibril protein to be lysozyme (Fig. 1).

Radiolabelled serum amyloid P component (SAP) scintigraphy

By the time of diagnosis, all symptomatic patients had amyloid in the liver and spleen and a large total body amyloid load by SAP scintigraphy. The kidneys were obscured by hepatosplenomegaly on the planar images in five cases. Amongst the remaining cases, the kidneys were abnormal in each of six with renal dysfunction and two of five without renal dysfunction. The asymptomatic patient (Patient 8, Table 1) had unequivocal amyloid in the spleen and kidneys by SAP scintigraphy but none in the liver. This patient did not have any tissue biopsies. Interestingly, amyloid did not continue to accumulate within native organs despite more than 8 years follow-up in four patients, contrasting recurrence of amyloid within three transplanted livers over a shorter time period.

DNA analysis and genetics

Twelve patients were heterozygous for a mutation in the lysozyme gene that encodes the D67H variant, two cases carried the W64R variant and one case each carried the I56T and D67G variants, all of which have previously been reported. Four kindreds accounted for 11 of 12 patients with the D67H variant (Table 1).

A family history of ALys was evident in every patient in the cohort. All except patient six had evidence of a 1st and 2nd degree relative with biopsy-proven amyloid or characteristic features of ALys. Patient 6 had a 2nd degree relative with amyloid and is a 4th degree relative of Patients 4 and 5. The highly penetrant nature of ALys is highlighted in the family tree of Patient 14 (Fig. 2). The proband from this family developed sicca syndrome aged 13 years. She had recurrent friable lesions on oesophagogastroduodenoscopy which presented as abdominal pain and haemorrhage and, aged 30 years, underwent an emergency small intestinal resection for spontaneous intestinal perforation. Two of the proband's sisters (Fig. 2, B_{VI} and B_{VIII}) also had endoscopically confirmed friable lesions in the stomach, which presented with early satiety and haematemesis, respectively. The proband's nephew (Fig. 2, C_{IV}) developed abdominal pain secondary to peptic ulceration at the age of 14 years. Two first cousins of the proband (Fig. 2; B_{IV} and B_{II}) developed anaemia on a background of overt GI haemorrhage in their third decade, which was associated, respectively, with oesophageal haemorrhagic nodules and gastric friable lesions.

Clinical course and outcome

Endoscopic investigation during the course of the illness was prompted by GI haemorrhage in six (38%) patients (Fig. 3), by dyspepsia in two cases and by nausea and vomiting in one case. Nine of 10 patients who underwent oesophagogastroduodenoscopy had endoscopic abnormalities such as peptic ulceration (four cases), oesophageal or gastric nodules and haemorrhagic gastritis. Contact bleeding was ob-

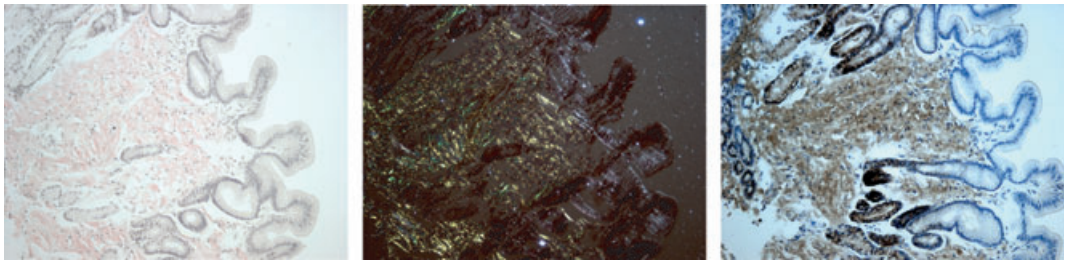


Fig. 1 *Gastric biopsy stained with Congo red showing extensive amyloid deposition; amorphous eosinophilic material viewed under brightfield light (left) and presence of red-green birefringence under cross-polarized light (middle). Immunohistochemical staining of the same biopsy with a monoclonal antibody directed against human lysozyme confirming lysozyme as the amyloid fibril protein (right).*

13652796, 2012, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2796.2011.02470.x by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [26/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

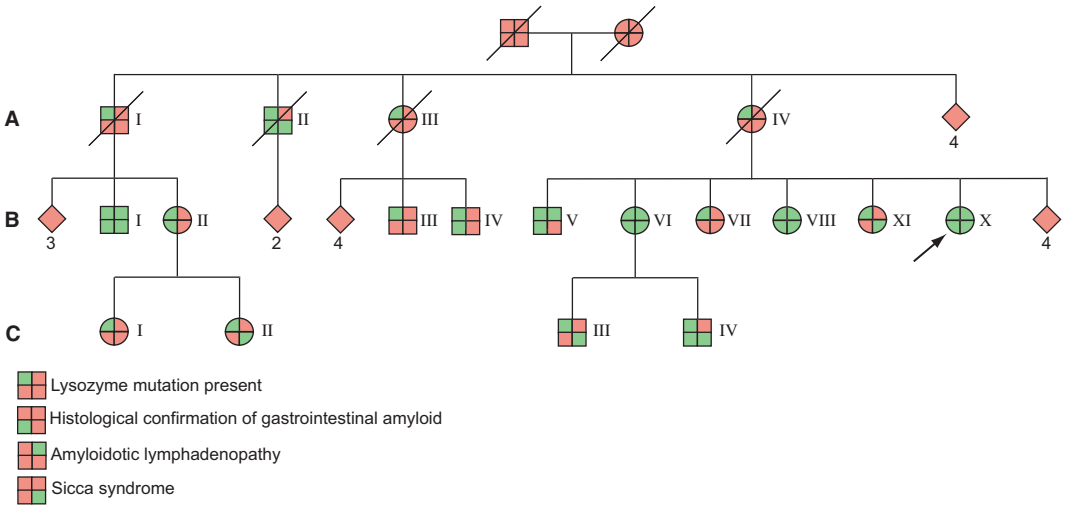


Fig. 2 *Family tree of Patient 14 with lysozyme amyloidosis (ALys). The proband is indicated by the arrow. Circles represent females and squares represent males. The diamonds with a number below them represent the number of unaffected siblings who are either asymptomatic or negative for the ALys mutation. Dead individuals are indicated by a diagonal line through the symbol.*

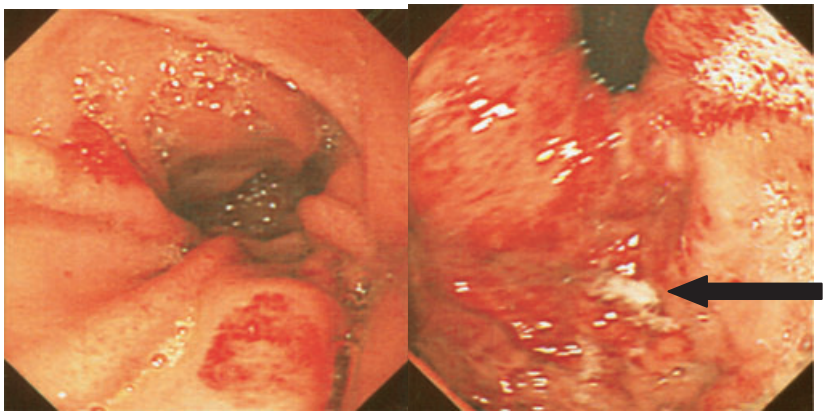


Fig. 3 *Oesophagogastroduodenoscopic views of the stomach of Patient 4 with lysozyme amyloidosis after presentation with upper gastrointestinal tract haemorrhage showing antral gastritis (left) and haemorrhagic gastritis of the fundus with ulceration (arrow).*

served on several endoscopic evaluations, even at points where the mucosa looked macroscopically normal and in the absence of clotting or platelet abnormalities. The four patients with peptic ulceration all received proton pump inhibitor (PPI) therapy although whether this was clinically beneficial is not clear; at least one patient had further GI bleeding whilst on chronic PPI therapy. Several patients (Table 1; Patients 6, 9 and 11) had diarrhoea associated with weight loss, and another (Patient 14) had a spontaneous small intestinal perforation which required resection and subsequent intestinal anastomosis, suggesting fragility of the intestinal wall from amyloid infiltration. Patient 6 had steatorrhoea with a faecal fat concentration of 60 mmol L^{-1} (normal <18 mmol L^{-1}) and subsequently presented with intestinal perforation secondary to a combination of diverticular disease and extensive GI amyloid infiltration, which required a Hartmann's procedure.

Despite extensive hepatic infiltration by amyloid by the time of diagnosis, liver function tests remained relatively preserved with a normal bilirubin in all patients and mildly elevated alkaline phosphatase [median (range) 77 (62–295) u L^{-1}] and aspartate aminotransferase values in only two and four cases, respectively. Median (range) serum albumin at diagnosis was 43 (27–50) g L^{-1}. Three patients (Table 1; Patients 4, 5 and 12) spontaneously ruptured their livers aged 24, 15 and 34 years,

 © 2011 The Association for the Publication of the Journal of Internal Medicine

13652796, 2012, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2796.2011.02470.x by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [26/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

respectively; each underwent emergency orthotopic liver transplantation (OLT) accompanied by perioperative splenectomy because of splenic fragility. The remaining OLT recipient (Patient 15) underwent the transplant because she had heavy hepatic amyloid infiltration and a strong family history (mother and sister) of spontaneous liver rupture. Patients 4 and 5 were siblings, and there had been a history of spontaneous hepatic rupture in both their affected father and paternal grandfather. The mother of patient 12 also had spontaneous hepatic rupture.

At censor, there were no hepatic graft failures 1.7, 5.8, 9.0 and 11.0 years from OLT. Serial SAP scintigraphy, however, showed asymptomatic amyloid recurrence within approximately 3 years in two patients (Table 1; Patients 4 and 5; Fig. 4), and amyloid was present in the graft of a third OLT recipient (Patient 12) by the time she first presented to our unit 7.8 years after the procedure. There was no evidence of amyloid recurrence in the remaining patient at censor 1.7 years after OLT.

Ten cases presented with or developed sicca syndrome, because of presumed salivary gland infiltration by amyloid. Sicca syndrome frequently predated the diagnosis of amyloidosis by several (range 3–30) years. 'Soft tissue' amyloid infiltration also manifested as petechiae in five patients and lymphadenopathy in two patients (Table 1; Patients 5 and 14); the latter was widespread in Patient 14 and was similarly so in three affected siblings and a third degree relative (Fig. 2).

Amyloidotic renal dysfunction occurred in seven patients during the course of their illness and was progressive culminating in end-stage renal disease (ESRD) in five such cases (Table 1; Patients 7, 9, 10, 11 and 16). Median (range) 24-h urinary protein loss amongst those presenting with renal dysfunction was only 0.3 g (0–4.0 g); a single patient had nephrotic-range proteinuria but was not clinically nephrotic. The remaining nine patients did not develop renal dysfunction during follow-up. Median time to ESRD from discovery of renal dysfunction by Kaplan–Meier analysis was 11.0 years.

Living donor renal transplantation (RTx) was undertaken in three patients (Table 1; Patients 9, 10 and 16), pre-emptively in two cases and 0.2 years after ESRD in patient 10. All three patients were alive at censor 6.6, 0.8 and 1.8 years after RTx, respectively, without evidence in any graft of amyloid recurrence. RTx is also known to have taken place in two relatives with ALys of patients in this cohort without evidence of graft failure after 8 years (brother of Patient 10) and 8 months (niece of Patient 14) from RTx.

Discussion

The hereditary non-neuropathic systemic amyloidosis syndromes are characterized by an autosomal dominant mode of inheritance and are typically associated with variable penetrance. Hereditary fibrinogen A-alpha chain and apolipoprotein AII amyloidosis cause renal failure, apolipoprotein AI amyloidosis is associated with renal, hepatic, cardiac, and laryngeal disease and gelsolin amyloidosis with cor-

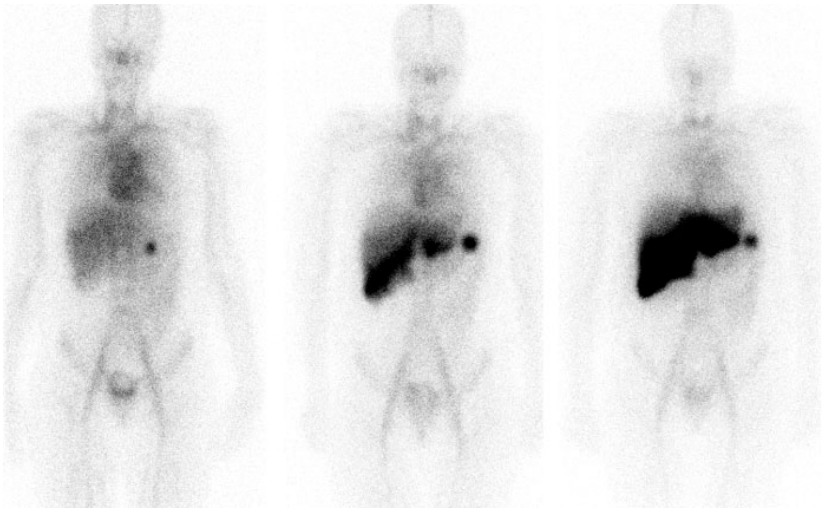


Fig. 4 *Serial anterior whole body ^{123}I-SAP scintigraphy of Patient 4 with lysozyme amyloidosis showing recurrence of amyloid in the liver graft after orthotopic liver transplantation (OLT) for hepatic rupture associated with progressive accumulation of amyloid within a splenunculus. The images were taken 5 (left), 41 (middle) and 65 (right) months after OLT.*

© 2011 The Association for the Publication of the Journal of Internal Medicine

13652796, 2012, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2796.2011.02470.x by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [26/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

neal lattice dystrophy, cranial neuropathy and occasionally, renal failure [12]. ALys is the most clinically heterogeneous condition in this group and may mimic the more common AL amyloidosis associated with plasma cell dyscrasias [13]. ALys is very penetrant, however, and a family history of amyloidosis is usually present in marked contrast to other forms of hereditary systemic amyloidosis [14, 15].

Distinguishing between ALys and AL amyloidosis is critical for correct patient management. First, the median estimated survival from diagnosis of amyloidosis in this series was 17.9 years, and despite median follow-up from symptom onset of 24.7 years, only four of 15 patients died (one patient remained asymptomatic throughout her disease course and was therefore excluded from the latter analysis). This contrasts systemic AL amyloidosis in which the median survival with modern chemotherapy regimens is approximately 3–4 years from diagnosis. In fact, median survival in the absence of chemotherapy amongst patients with AL amyloidosis is approximately 1 year, whereas there is no role for chemotherapy in ALys. One patient in this series (Table 1; Patient 7) received inappropriate chemotherapy for presumed AL amyloidosis prior to review in our unit. It is imperative, therefore, that patients with systemic amyloidosis are screened for a family history of similar disease and undergo genetic sequencing, particularly if the amyloid fibril protein remains uncertain after immunohistochemical staining of amyloidotic tissue, which may be the case in up to 40% of patients with AL amyloidosis [14]. More recently, mass spectrometry has emerged as a sensitive technique for identifying amyloid fibril proteins [16].

Symptomatic GI amyloid is relatively unusual in the other forms of hereditary non-neuropathic systemic amyloidosis but was prevalent in this ALys cohort. GI haemorrhage associated with endoscopically visible, friable lesions was common in this series and has previously been reported in a French family with ALys [17]. Our practice has been to administer lifelong prophylactic PPI therapy in all patients with ALys who have GI amyloid to try and reduce risk of upper GI haemorrhage. Whether this practice is clinically beneficial, however, cannot be concluded from this case series. Three patients in our series had severe diarrhoea and weight loss including one with fat malabsorption suggestive of small intestinal amyloid, which has also been reported previously in ALys [18].

Hepatic amyloid, detected by SAP scintigraphy, was virtually universal at diagnosis in these patients, along with a large total body amyloid burden. Interestingly, liver function tests were barely deranged and remained relatively stable during follow-up, which contrasts AL amyloidosis, in which patients with similar SAP scintigraphic appearances have markedly deranged liver biochemistry and may develop hepatic decompensation [19]. Both this and the finding that median time from presentation with renal dysfunction to ESRD in this cohort was 11.0 years indicate that the natural history of ALys is markedly slower than AL and other types of hereditary amyloidosis [15, 20]. The mechanism by which amyloid accumulation in native organs appears to abate in ALys once patients have accumulated a large total body amyloid burden remains unclear.

Hepatic haemorrhage and/or rupture occurred in three of 12 patients with ALys associated with the D67H variant in this series. As the diagnosis of amyloid was only established after liver rupture in all three of these cases, it may be that the prevalence suggested here is falsely high. Patient 5's explanted liver was noted to be fragile and friable with loss of the reticulin framework [21] and that of Patient 4 was 3 kg, which is approximately twice the weight of a normal liver [22]. Although the extent of amyloid infiltration may contribute to risk of haemorrhage, it is notable that liver rupture in AL amyloidosis, which is also associated with massive hepatomegaly, is exceptionally rare. Emergency OLT amongst these three patients was life-saving and appears to have a good outcome in terms of graft survival, despite evidence of amyloid recurrence in the grafts, with 100% 1- and 5-year graft survival in our small series. One of the cohort had a 'prophylactic' OLT on a background of extensive hepatic amyloidosis and a family history of hepatic rupture. In a climate of donor shortages, the merits of such an approach remain uncertain. We would, however, recommend a pre-emptive OLT assessment in those with confirmed ALys particularly if there is a family history of hepatic rupture or haematoma.

Soft-tissue amyloidosis, one manifestation of which is sicca syndrome, is recognized in both ALys and AL amyloidosis [23, 24]. Sicca syndrome frequently predated vital organ dysfunction in this cohort and should prompt a search for amyloid deposits in any patient with a family history of ALys. Indeed, two patients in this series were diagnosed in precisely this fashion. Artificial saliva or tears may provide symptomatic relief. Other 'soft-tissue' manifestations of amyloid in this cohort were bruising and lymph node enlargement, which are usually considered to be

13652796, 2012, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2796.2011.02470.x by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [26/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

almost pathognomonic of AL amyloidosis [13]. Patient 14 had evidence of widespread amyloidotic lymphadenopathy, which has not been reported before in ALys.

Renal dysfunction is common in all forms of hereditary non-neuropathic systemic amyloidosis, and five of 16 cases in this series reached ESRD. Age at presentation and rate of renal progression varied widely, even amongst patients from the same family who carried the same mutation, evident from Family 4 in this series and consistent with previous reports [4, 5, 25]. Despite a large total body amyloid burden and a complete lack of available therapy to halt amyloid accumulation in ALys, outcomes amongst three RTx recipients were excellent with functioning renal allografts and no recurrence of amyloid 0.8, 1.8 and 6.6 years after RTx. To our knowledge, there has never been a report of amyloid-related renal allograft failure in ALys despite ongoing production of amyloidogenic precursor protein [3, 26, 27].

Conclusion

Lysozyme amyloidosis is a rare form of hereditary non-neuropathic systemic amyloidosis, which is clinically heterogeneous and may present with GI haemorrhage, hepatic rupture, sicca syndrome, purpura and petechiae, lymphadenopathy or renal dysfunction. There is considerable overlap between the clinical features in ALys and AL amyloidosis, and it is important to distinguish between these diagnoses as there is no role for chemotherapy in ALys. Clinical penetrance in the presence of an amyloidogenic lysozyme mutation appears to be high, evident by the obvious family history in all cases within this cohort. Although there is no existing amyloid-specific therapy for ALys, which is therefore managed symptomatically, the condition has a slow natural history and it is even possible that amyloid accumulation may abate in some patients. OLT for cases of liver rupture and RTx for ESRD appear to be very successful with excellent medium-term graft function and patient survival.

Acknowledgements

This work was supported by funding from the UK Department of Health, MRC Programme Grant G97900510 to P.N.H. and the UCL Amyloidosis Research Fund. We would like to also acknowledge all the physicians and surgeons who were involved in the clinical care of the patients reported in this study.

Conflict of interest statement

No conflicts of interest to declare.

References

1 Ostertag B. Demonstration einer eigenartigen familiaren paraamyloidose. *Zentralbl Aug Pathol* 1932; **56:** 253–4.
2 Pepys MB, Hawkins PN, Booth DR *et al.* Human lysozyme gene mutations cause hereditary systemic amyloidosis. *Nature* 1993; **362:** 553–7.
3 Yazaki M, Farrell SA, Benson MD. A novel lysozyme mutation Phe57Ile associated with hereditary renal amyloidosis. *Kidney Int* 2003; **63:** 1652–7.
4 Valleix S, Drunat S, Philit JB *et al.* Hereditary renal amyloidosis caused by a new variant lysozyme W64R in a French family. *Kidney Int* 2002; **61:** 907–12.
5 Wooliver C, Coriu D, Murphy CL *et al.* Familial amyloidosis associated with a novel mutation (D68G) in the lysozyme gene. XIth International Symposium on Amyloidosis 2008: 208–10.
6 Rocken C, Becker K, Fandrich M *et al.* ALys amyloidosis caused by compound heterozygosity in exon 2 (Thr70Asn) and exon 4 (Trp112Arg) of the lysozyme gene. *Hum Mutat* 2006; **27:** 119–20.
7 Booth DR, Pepys MB, Hawkins PN. A novel variant of human lysozyme (T70N) is common in the normal population. *Hum Mutat* 2000; **16:** 180.
8 Puchtler H, Sweat F, Levine M. On the binding of Congo red by amyloid. *J Histochem Cytochem* 1962; **10:** 355–64.
9 Tennent GA. Isolation and characterization of amyloid fibrils from tissue. In: Wetzel R, ed. *Methods in Enzymology: Amyloid, Prions and Other Protein Aggregates*. San Diego, CA: Academic Press Ltd, 1999; 26–47.
10 Talmud P, Tybjaerg-Hansen A, Bhatnagar D *et al.* Rapid screening for specific mutations in patients with a clinical diagnosis of familial hypercholesterolaemia. *Atherosclerosis* 1991; **89:** 137–41.
11 Hawkins PN, Lavender JP, Pepys MB. Evaluation of systemic amyloidosis by scintigraphy with ^{123}I-labeled serum amyloid P component. *N Engl J Med* 1990; **323:** 508–13.
12 Benson MD. Ostertag revisited: the inherited systemic amyloidoses without neuropathy. *Amyloid* 2005; **12:** 75–87.
13 Wechalekar AD, Hawkins PN, Gillmore JD. Perspectives in treatment of AL amyloidosis. *Br J Haematol* 2008; **140:** 365–77.
14 Lachmann HJ, Booth DR, Booth SE *et al.* Misdiagnosis of hereditary amyloidosis as AL (primary) amyloidosis. *N Engl J Med* 2002; **346:** 1786–91.
15 Gillmore JD, Lachmann HJ, Rowczenio D *et al.* Diagnosis, pathogenesis, treatment, and prognosis of hereditary fibrinogen A alpha-chain amyloidosis. *J Am Soc Nephrol* 2009; **20:** 444–51.
16 Vrana JA, Gamez JD, Madden BJ, Theis JD, Bergen HR III, Dogan A. Classification of amyloidosis by laser microdissection and mass spectrometry-based proteomic analysis in clinical biopsy specimens. *Blood* 2009; **114:** 4957–9.
17 Granel B, Serratrice J, Valleix S *et al.* A family with gastrointestinal amyloidosis associated with variant lysozyme. *Gastroenterology* 2002; **123:** 1346–9.
18 Zalin AM, Jones S, Fitch NJS, Ramdsden DB. Familial nephropathic non-neuropathic amyloidosis: clinical features, immunohistochemistry and chemistry. *Q J Med* 1991; **81:** 945–56.
19 Sattianayagam P, Gibbs SD, Wechalekar AD *et al.* Hepatic AL amyloidosis – a 10-year experience at the United Kingdom

© 2011 The Association for the Publication of the Journal of Internal Medicine

13652796, 2012, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2796.2011.02470.x by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [26/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

National Amyloidosis Centre (NAC). *Gastroenterology* 2010; **138 (Suppl. 1):** S-366.
20 Gillmore JD, Stangou AJ, Lachmann HJ *et al.* Organ transplantation in hereditary apolipoprotein AI amyloidosis. *Am J Transplant* 2006; **6:** 2342–7.
21 Harrison RF, Hawkins PN, Roche WR, MacMahon RFT, Hubscher SG, Buckels JAC. 'Fragile' liver and massive hepatic haemorrhage due to hereditary amyloidosis. *Gut* 1996; **38:** 151–2.
22 Mells GF, Buckels JA, Thorburn D. Emergency liver transplantation for hereditary lysozyme amyloidosis. *Liver Transpl* 2006; **12:** 1908–9.
23 Prokaeva T, Spencer B, Kaut M *et al.* Soft tissue, joint, and bone manifestations of AL amyloidosis: clinical presentation, molecular features, and survival. *Arthritis Rheum* 2007; **56:** 3858–68.
24 Simon BG, Moutsopoulos HM. Primary amyloidosis resembling sicca syndrome. *Arthritis Rheum* 1979; **22:** 932–4.
25 Gillmore JD, Booth DR, Madhoo S, Pepys MB, Hawkins PN. Hereditary renal amyloidosis associated with variant lysozyme in a large English family. *Nephrol Dial Transplant* 1999; **14:** 2639–44.
26 Lanham JG, Meltzer ML, de Beer FC, Hughes GRV, Pepys MB. Familial amyloidosis of Ostertag. *QJ Med* 1982; **51:** 25–32.
27 Granel B, Valleix S, Serratrice J *et al.* Lysozyme amyloidosis: report of 4 cases and a review of the literature. *Medicine (Baltimore)* 2006; **85:** 66–73.

Correspondence: Dr Julian D. Gillmore, National Amyloidosis Centre, CAAPP, UCL Medical School, Royal Free Hospital, Rowland Hill Street, London NW3 2PF, UK.
(fax: +44 0 20 7433 2817; e-mail: j.gillmore@ucl.ac.uk).

 © 2011 The Association for the Publication of the Journal of Internal Medicine

13652796, 2012, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2796.2011.02470.x by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [26/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License